

SpatialLens Assist: Motion-Aware Hazard Detection for Campus Navigation

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Abstract

Assistive navigation systems must communicate not only *what* is nearby but *how* it moves relative to the walker. We present **SpatialLens Assist**, an offline pipeline that converts short campus-walkway videos into motion-aware hazard alerts for blind and low-vision pedestrians. Each tracked object is classified into one of six hazard categories using a rule cascade over ego-motion-compensated optical flow and bounding-box growth, with a textual evidence string for every prediction. On 11 campus videos and 61 manually labeled tracks, the system achieves **93.3% decidable accuracy** on tracks with ≥ 3 frames and **82.5% overall accuracy**; the gap is dominated by deliberate **short_track** abstentions on 1–2 frame tracker fragments. The safety-critical **approaching** class reaches precision 0.95 and recall 0.88. Ablations show that raising the tracker frame-gap buffer (+15.3 pp) and a v2 rule cascade (+3.6 pp vs. removing **strong_crossing**) are the largest accuracy drivers; an opt-in ByteTrack backend is a negative result for low-fps offline footage.

1 Introduction

Camera-based assistive navigation has progressed rapidly through object detectors that can localize pedestrians, bicycles, and other road users in real time. For a blind pedestrian, however, knowing that a bicycle is present in the frame is insufficient: the relevant question is whether it is *approaching*, *crossing the path*, or *moving away*. Single-frame appearance cues are ambiguous—a distant cyclist and a parked bicycle can look similar—while motion over time disambiguates intent.

SpatialLens Assist targets this gap with a lightweight, explainable pipeline that runs on iPhone campus footage sampled at 2 fps. The system detects objects, tracks them across frames, extracts motion features with ego-motion compensation, classifies each track into a hazard label, and emits natural-language alerts that quote the classifier’s evidence string.

1.1 Problem Statement

Given a short egocentric video from a handheld phone, produce for each persistent object track: (1) a hazard label from {**approaching**, **crossing_left_to_right**, **crossing_right_to_left**, **moving_away**, **static**, **uncertain**}; (2) a confidence tier; and (3) a human-readable alert suitable for text-to-speech output. The system must abstain honestly when motion evidence is insufficient rather than guess.

1.2 Contributions

This report documents a complete week-1–3 pipeline with: (i) ego-motion-compensated motion features for hand-held footage; (ii) a priority-ordered rule cascade with explainable **uncertain_reason** sub-labels; (iii) a 61-track manually labeled evaluation set across 11 campus videos; (iv) ablation studies on tracker frame-gap, classifier cascade design, and ByteTrack; and (v) reproducible scripts, 80 unit tests, and slide-ready demo assets.

2 Related Work

Object detection for assistive navigation commonly builds on YOLO-family detectors [1] applied to egocentric video. Multi-object tracking for offline clips often uses IoU matching or Kalman-filter trackers such as ByteTrack [2]; the latter is designed for high-fps real-time streams and may discard low-confidence detections that are common at 2 fps sampling. Motion analysis for collision warning traditionally combines optical flow [3] with heuristics on bounding-box scale change and image-plane displacement. Learned trajectory classifiers require labeled training data that was unavailable at project start; we therefore adopt an interpretable rule cascade whose thresholds can be tuned against a small hand-labeled set without overfitting.

3 Methods

Figure 1 summarizes the end-to-end architecture. The pipeline has five stages: detection, tracking, motion-feature extraction, hazard classification, and alert generation.

3.1 Detection and Tracking

Detection. Ultralytics YOLOv8n [1] on COCO classes {person, bicycle, motorcycle, skateboard} with confidence threshold $\tau = 0.35$. Scooters are not a native COCO class and may appear as `skateboard`.

Tracking. A per-class IoU+centroid matcher links detections across frames using score $0.7 \cdot \text{IoU} + 0.3(1 - \hat{d})$, where $\hat{d}$ is normalized centroid distance. Unmatched detections spawn new track IDs; unmatched tracks are kept for up to `max_frame_gap=4` frames before deletion. An opt-in ByteTrack backend was implemented but not adopted (Section 6).

3.2 Motion Feature Extraction

For each track we compute: (i) start-to-end centroid displacement (dx, dy) and normalized horizontal displacement dx_norm ; (ii) bounding-box growth ratio $\text{growth} = (A_{\text{end}} - A_{\text{start}})/A_{\text{start}}$; (iii) median Farneback optical-flow magnitude and direction inside the bbox; and (iv) frame-differencing overlap within the bbox.

Both flow and frame-diff are *ego-motion compensated*: the global median flow vector is subtracted from per-object flow, and consecutive frames are aligned with ECC (translation model) before differencing. This reduces false motion from hand-held camera pans.

3.3 Hazard Rule Cascade

The classifier applies a fixed priority cascade. Tracks with fewer than `min_track_frames=3` abstain as `uncertain/short_track`. Otherwise rules fire in order: (1) `moving_away` if $\text{growth} < -0.15$; (2) `strong_crossing` if $|dx_norm| > 0.50$ (v2 addition); (3) `crossing_*` if $|dx_norm| > 0.12$ and growth is below threshold; (4) `approaching` if growth and `approach_score` exceed thresholds; (5) `static` if displacement and flow are near zero; else (6) `uncertain` with a reason tag (`near_approaching`, `low_signal`, etc.). Every label includes a textual evidence string used by the alert layer.

3.4 Assistive Alerts and Explainability

The alert module maps hazard labels to spoken messages (e.g. “Bike approaching from the left.”). For `uncertain` tracks the message quotes the `uncertain_reason`: “Person detected briefly; not enough frames to judge motion” for `short_track`. Annotated frames and MP4 demos overlay bbox color, hazard label, and evidence text;

videos are encoded with H.264 for macOS QuickTime compatibility.

4 Dataset and Evaluation Protocol

Videos. Eleven *controlled* campus-walk clips (IMG_4972–4982) filmed on Stanford walkways at 2fps; two additional *uncontrolled* natural-walk clips (IMG_4989–4990) for qualitative demo only.

Labels. Ground truth in `data/labels/hazard_labels.csv`: 61 tracks manually labeled with hazard category and reviewer notes. Four labeled tracks were merged by the tracker and appear as 57 evaluated prediction–label pairs.

Metrics. For matched tracks we report:

- (i) **Overall accuracy:** fraction of tracks where predicted label equals ground truth.
- (ii) **Decidable accuracy:** same, restricted to tracks with ≥ 3 frames—the regime the motion classifier is designed for.
- (iii) **Per-class precision, recall, F1** and macro-F1.
- (iv) **Confusion matrix** (rows = true, columns = predicted).

5 Results

5.1 Quantitative Results

Table 1 reports per-class metrics on the aggregate labeled set. Table 2 summarizes ablations. Headline numbers: **93.3 % decidable** (28/30), **82.5 % overall** (47/57), macro-F1 0.671. The safety-critical `approaching` class—the primary collision-warning category—achieves F1 0.91 at precision 0.95.

Table 1: Per-class metrics on 57 evaluated tracks (11 videos).

Class	Prec.	Rec.	F1	<i>n</i>
approaching	0.95	0.88	0.91	24
moving_away	1.00	0.78	0.88	9
crossing_R→L	1.00	0.67	0.80	3
uncertain	0.63	1.00	0.77	15
crossing_L→R	1.00	0.50	0.67	4
static	0.00	0.00	0.00	2
Macro F1	0.671			
Decidable acc. (≥ 3 fr.)	93.3 % (28/30)			
Overall accuracy	82.5 % (47/57)			

5.2 Failure Analysis

Figure 2 decomposes remaining errors. **Confusion matrix (a):** the `approaching` column is clean; most off-diagonal mass flows to `uncertain` on 1–2 frame tracker

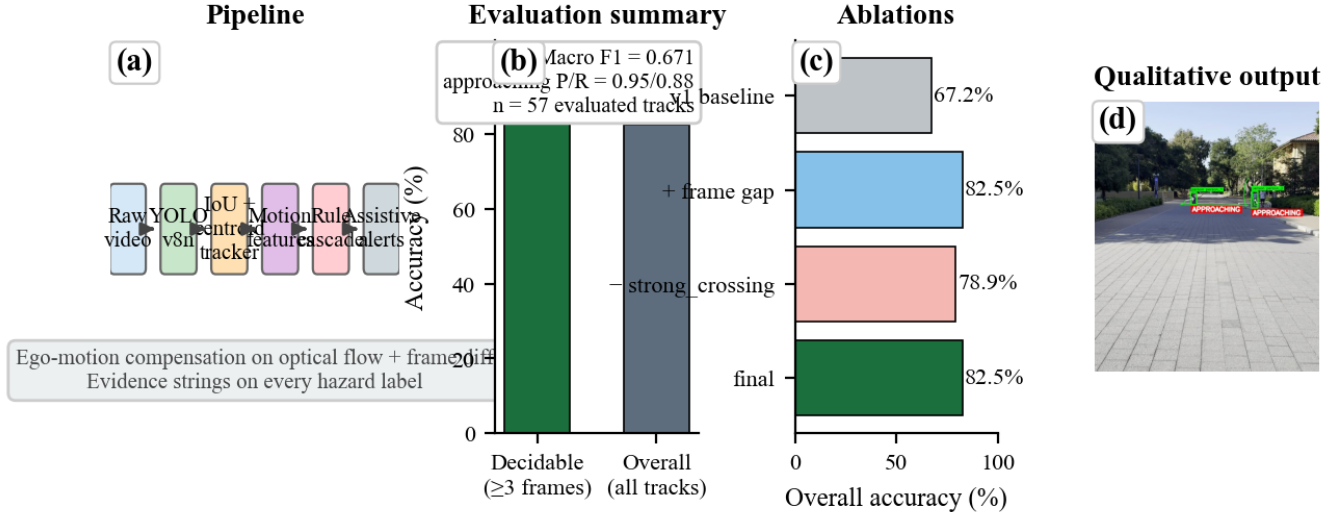


Figure 1: Overview of SpatialLens Assist. (a) End-to-end pipeline. (b) Decidable vs. overall accuracy on 57 matched tracks. (c) Ablation accuracies. (d) Qualitative hazard overlay from IMG_4976.

Table 2: Ablation study on the labeled evaluation set.

Configuration	Accuracy
v1 baseline (<code>max_frame_gap=2</code>)	67.2 %
+ frame gap ($2 \rightarrow 4$)	82.5 %
– <code>strong_crossing</code> rule (v2 cascade)	78.9 %
Final (v2 cascade + <code>gap=4</code>)	82.5 %
ByteTrack backend (tuned, low fps)	~50 % det. retained

fragments. **Uncertain reasons (b):** `short_track` accounts for 98 % of uncertain predictions (39/40), indicating tracker fragmentation rather than classifier confusion as the dominant failure mode. **Priority conflicts** (diagonal walkers labeled `approaching` instead of `crossing`) were reduced by the v2 cascade; one `moving_away` track is still misclassified as `approaching` when bbox growth briefly spikes. **Static objects** (parked bicycles) appear for only one frame and correctly abstain as `short_track`, yielding 0 % static F1 on the labeled set.

5.3 Qualitative Results

Figure 3 shows representative outputs. Trajectories from IMG_4974 illustrate centroid paths used for crossing detection. The hazard overlay from IMG_4976 highlights approaching pedestrians with red **APPROACHING** banners and prints evidence strings below each bbox. Uncontrolled clip IMG_4989 exposes the remaining limit: ~50 % of tracks fall to `uncertain/short_track` when objects pass through the frame in only 1–2 sampled frames.

6 Discussion and Deviations

Rule cascade vs. learned classifier. The proposal hedged between learned and rule-based classification.

We committed to rules because no labeled set existed in week 1, rules produce explicit evidence strings, and thresholds tune against 61 labels without overfitting a deep model.

Ego-motion compensation. Added in week 3 after hand-held pans produced false-positive flow; median-flow subtraction + ECC alignment reduced spurious `approaching` labels.

ByteTrack (negative result). Implemented and benchmarked as a drop-in tracker replacement. Even tuned for 2 fps offline data, ByteTrack dropped roughly half the detections—its Kalman design assumes high-fps real-time streams. The simpler IoU+centroid matcher remains default.

Scope unchanged. Hazard taxonomy, campus-walkway setting, and assistive-alert framing match the proposal.

7 Remaining Work

With one week remaining: (**day 1**) optional post-hoc track merger (greedy per-class merge of fragments ≤ 3 frames apart, target 87–90 % overall); (**days 2–4**) lock numbers, methods chapter, failure analysis; (**days 5–6**) slide deck from `outputs/slide_assets/`; (**day 7**) buffer. Stretch goals (YOLOv8s on IMG_4977, slow-approach softening rule) are off the critical path; we will not re-sample or re-label, as that would invalidate the evaluation set. All code and configs are versioned; the pipeline runs end-to-end via `python scripts/run_final_pipeline.py --all`.

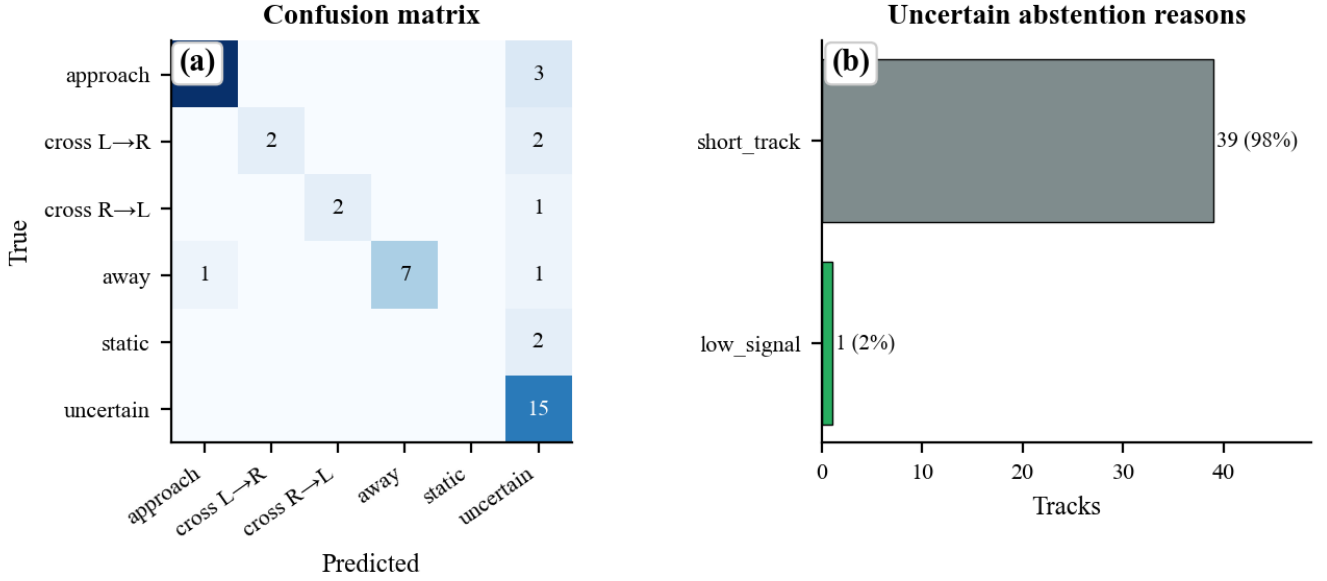


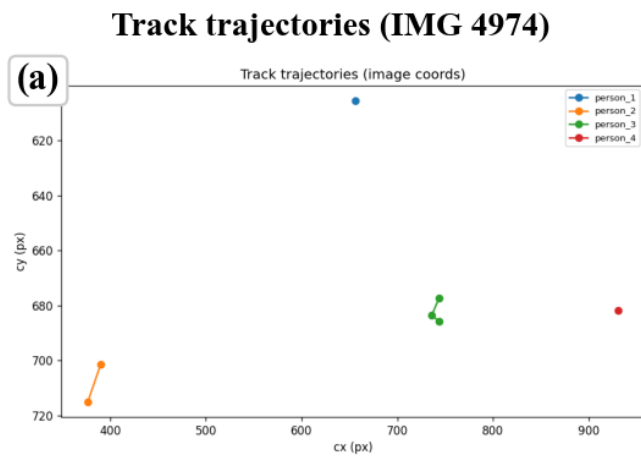
Figure 2: Evaluation and failure analysis. (a) Aggregate confusion matrix. (b) `uncertain_reason` breakdown; `short_track` dominates, motivating post-hoc track merging.

8 Conclusion

SpatialLens Assist demonstrates that motion-aware hazard classification for campus navigation is feasible with a lightweight, explainable pipeline. On tracks with sufficient temporal support, decidable accuracy reaches 93.3%; the remaining overall gap is largely tracker fragmentation, not classifier error. The v2 rule cascade and frame-gap tuning provide the largest gains; ByteTrack is a useful negative result for low-fps offline settings. Post-hoc track merging is the most promising next step before the final report and demo.

References

- [1] J. Redmon and A. Farhadi. YOLOv3: An incremental improvement. *arXiv:1804.02767*, 2018. Ultralytics YOLOv8 used in implementation.
- [2] Y. Zhang et al. ByteTrack: Multi-object tracking by associating every detection box. *ECCV*, 2022.
- [3] G. Farnebäck. Two-frame motion estimation based on polynomial expansion. *SCIA*, 2003.
- [4] G. Evangelidis and E. Psarakis. Parametric image alignment using enhanced correlation coefficient maximization. *IEEE TPAMI*, 2008.



Hazard overlay (IMG 4976)



Figure 3: Qualitative pipeline outputs. (a) Track trajectories in image coordinates (IMG_4974). (b) Hazard-annotated frame with evidence strings (IMG_4976).